Predicting Severity of Emerging Infectious Diseases Using Machine Learning

Course: Population Health Informatics

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Group Number: Group 3

Introduction to Emerging Infectious Diseases

What are Emerging Infectious Diseases?

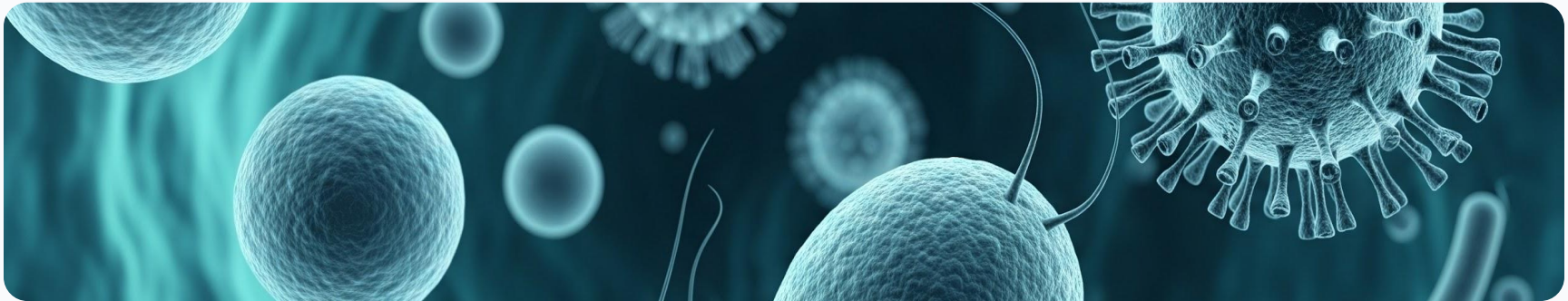
Diseases that are either **new** or **rapidly increasing** in a population.

Key Characteristics

- Newly identified in a population
- Rapidly increasing in incidence or geographic spread

Primary Causes

- **Viruses** (e.g., COVID-19)
- **Bacteria**
- **Parasites**



Re-emerging Infectious Diseases

Previously controlled diseases that are coming back, requiring advanced informatics to understand patterns and severity variations.

Reasons for Re-emergence

- Antibiotic resistance
- Reduced vaccination rates
- Environmental and climate changes

Key Examples

- Covid 19
- Ebola

Public Health Informatics: Leveraging CDC data to track illness patterns and improve response strategies.

Problem Statement

Hard to identify high-risk patients early
Early identification is critical to prevent severe outcomes and system strain.

Leads to:

- Delayed treatment
- Overwhelmed hospitals

The Solution

Data-Driven Prediction

Advanced informatics tools are required to forecast severity and prioritize care.

Action Required: Leveraging AI and machine learning to bridge the gap in early patient assessment.

Impact on Population Health

Early Detection of Outbreaks

AI & ML Analysis

Analyze large datasets with symptoms and demographic data to detect patterns early.

Faster Response

Detect outbreaks before traditional methods, promoting faster containment of infections.

Predicting Disease Severity

High-Risk Identification

Models identify high-risk patients early to prioritize hospital and ICU resource allocation.

Outcome Improvement

Leverage data-driven insights to significantly improve overall patient outcomes.

Impact on Population Health (Cont.)

Disease Surveillance

Real-time monitoring at health agencies like CDC, CSTE, NNDSS.

Track disease spread across geographic areas quickly.

Support data-driven decision making.

Drug & Vaccine R&D

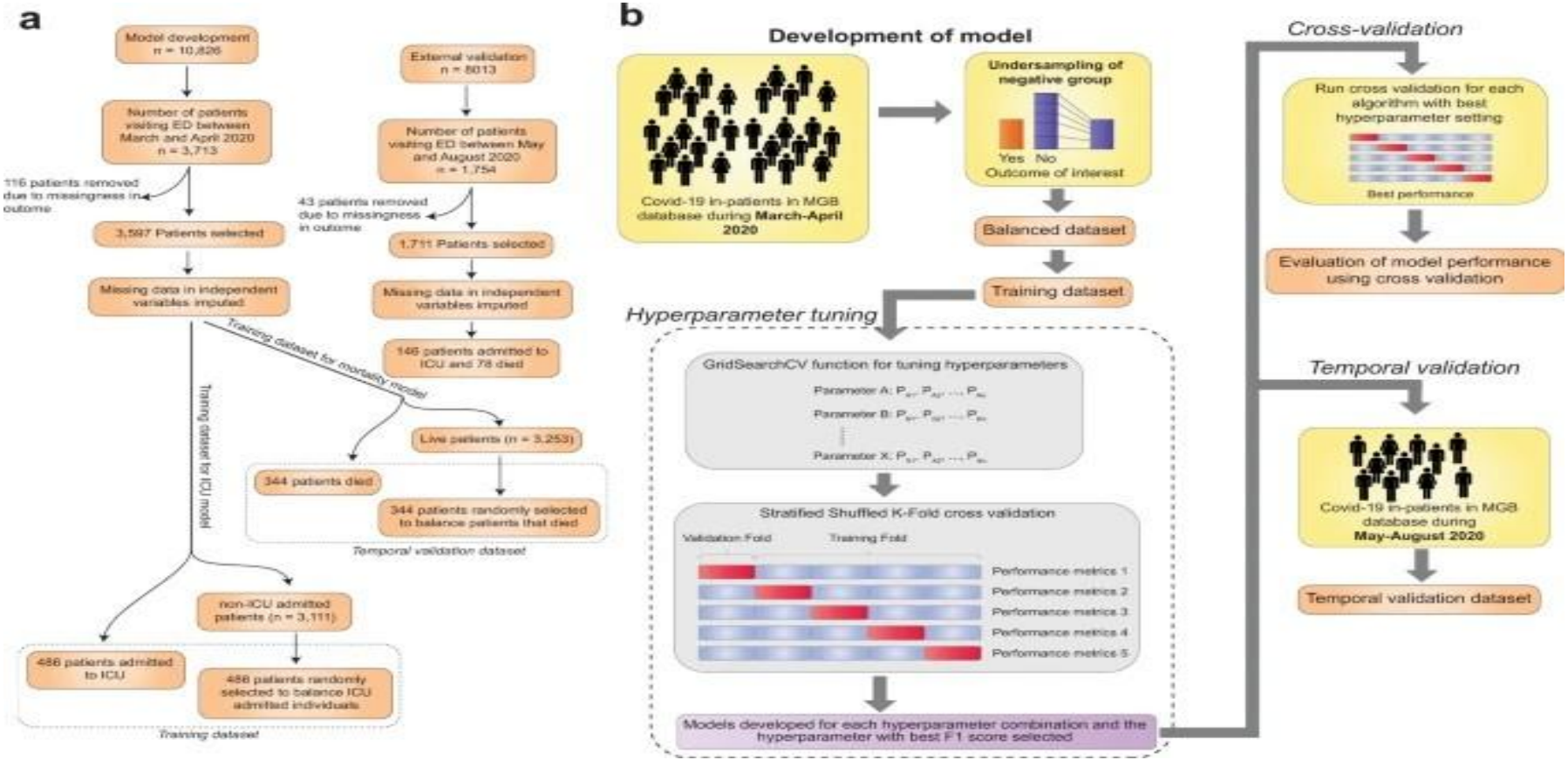
AI helps identify drug targets and speed up vaccine research.

Reduces time needed for development during pandemics.

Predictive Modelling

Used to forecast case numbers and peak infection periods.

Schematic diagram of ML model development



Hypothetical XGBoost Disease Severity Model

1. Define & Prepare

Define Outcome

Binary severity (0=non-severe, 1=severe: ICU, ventilation, death).

Collect Data

Demographics, vitals, labs, comorbidities; clean & encode.

2. Data Processing

Split Data

Train/test split (70/30) or cross-validation.

Class Imbalance

Apply `scale_pos_weight` or resampling.

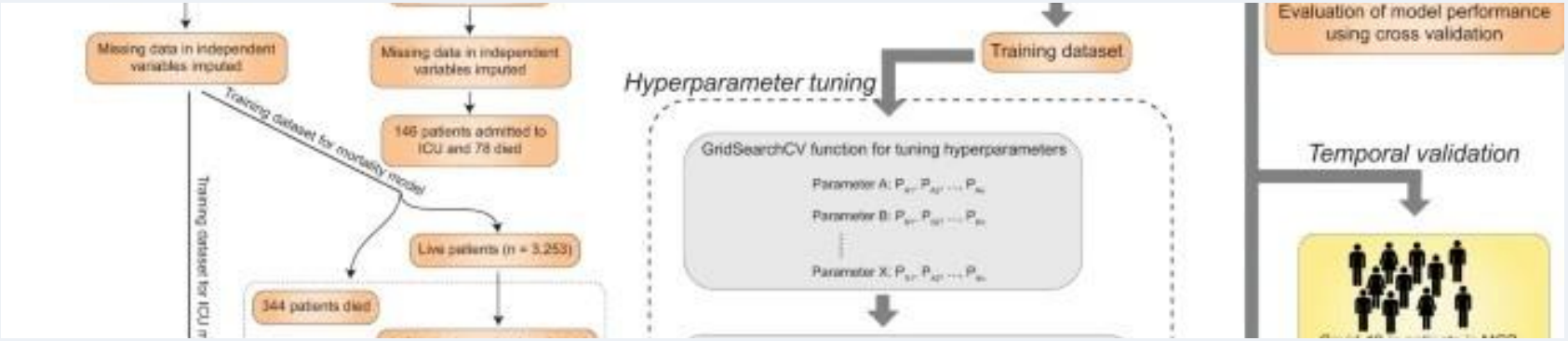
3. Model Training

Train XGBoost

Fit model using key predictors/features.

Hyperparameters

Optimize `max_depth`, `lr`, `n_estimators`.



Hypothetical XGBoost Disease Severity Model (Cont.)

4. Tune Hyperparameters

Optimize performance using parameters like max_depth, learning_rate, and n_estimators.

5. Evaluate Performance

Assess clinical utility focusing on metrics such as ROC-AUC, recall (sensitivity), and F1-score.

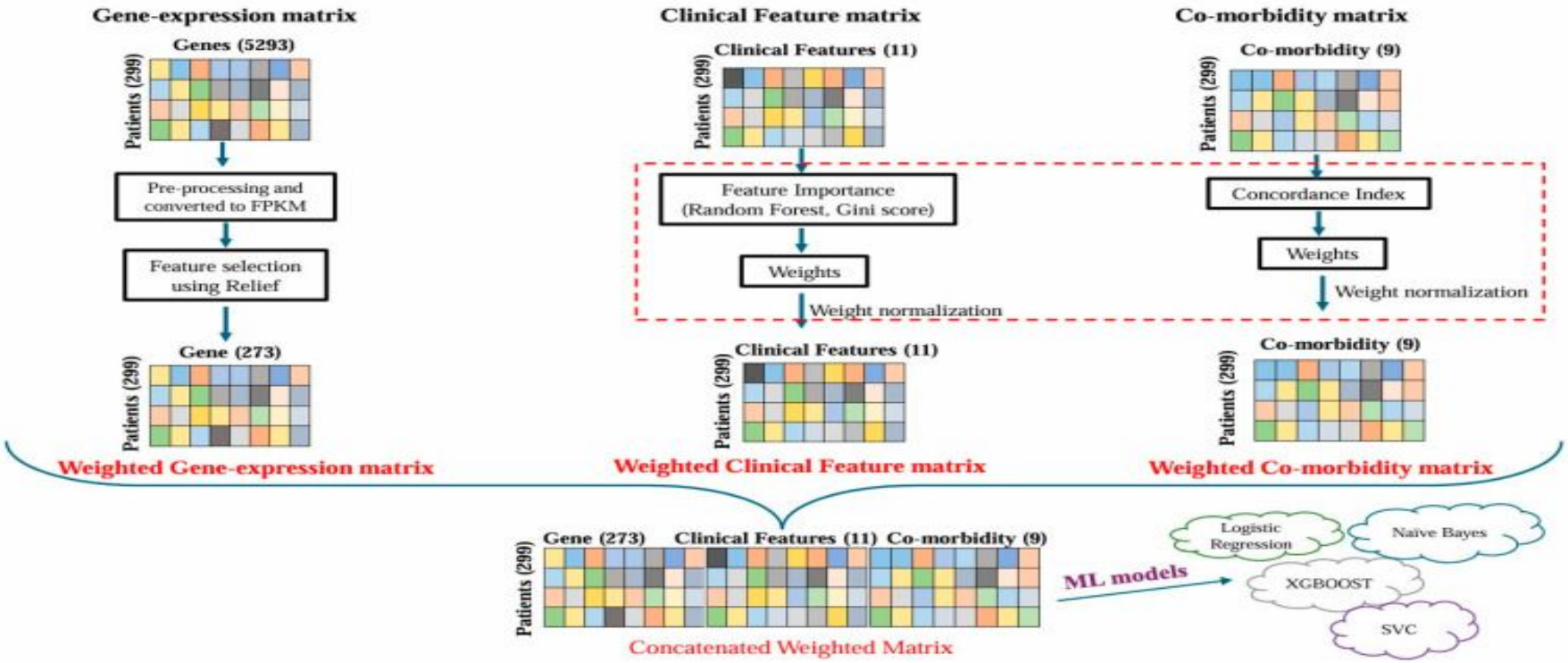
6. Interpret Model

Identify key predictors and understand decision logic using feature importance or SHAP values.

7. Validate & Deploy

Test on independent new data to ensure reliability for real-world clinical decision support.

Data processing workflow example



Workflow of preprocessing various data types, generating individual feature weight matrices and integrations, and machine learning model training for COVID-19 severity prediction (Sethi et al, 2024).

Key Features in Predictive Models

Key Indicators

Common features used to predict severity include:

- Absolute neutrophil count
- Viremia categories
- Co-morbidities

Feature Selection

Techniques like **LASSO regression** identify significant markers of disease progression.

Effectively distinguishes between critical and non-critical cases.

Model Interpretability

Tools like **SHAP** explain how specific features contribute to patient risk scores.

Essential for building trust in AI-driven clinical decisions.

Past Research Findings

ML has been used to predict disease severity and mortality in emerging infections.

SFTS Early Mortality Prediction

Study on Severe fever with thrombocytopenia syndrome (SFTS) used 8 ML models (RF, XGBoost, GBM, etc.) for early mortality prediction.

XGBoost Performance

Achieved favorable metrics: sensitivity (0.810), specificity (0.864), F1 score (0.905), accuracy (0.854), and AUC (0.91) (Zhao et el, 2026).

High-Risk Identification

Sethi et al. (2024) demonstrated models using clinical and laboratory data for early identification of high-risk patients.

Data Modality Integration

Integrating RNA-seq, clinical, and comorbidity data significantly improves COVID-19 severity prediction accuracy over single modalities.

Past Research Findings cont…

Surveillance & Prediction

ML can be used for disease surveillance, outbreak prediction, and early warning systems (Nasr et al, 2025).

Data Integration

AI models integrate clinical, environmental, and population data for real-time insights.

Digital Epidemiology

Tools using social media and travel data detect outbreaks earlier than traditional systems (Nasr, et al, 2025).

Severity Prediction

ML and DNNs predict severe EID illness using comorbidities and demographics (Chiu et al, 2022).

Key Impact

Improved preparedness, policy decisions, and pandemic response for emerging infectious diseases (EIDs).

Challenges Associated with Use of AI

Data Quality & Bias

Data quality affects performance; low-quality results in inaccurate predictions. Bias in datasets leads to biased outcomes.

Generalizability

Models developed for specific populations must be validated before being applied to others efficiently.

Data Privacy

Health data concerns limit access to large datasets required for effective model training.

Real-time Implementation

Most models are based on retrospective studies, making real-time clinical application challenging.

Conclusion

Data-Driven Public Health

Predicting the severity of emerging infectious diseases using machine learning demonstrates the powerful role of data-driven approaches in modern public health.

Early Risk Identification

By leveraging large-scale surveillance data from organizations like the Centers for Disease Control and Prevention, this project shows how early identification of high-risk patients is possible.

Accurate Risk Stratification

The use of models such as logistic regression or XGBoost enables more accurate risk stratification, supporting timely clinical intervention and efficient resource allocation.

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